

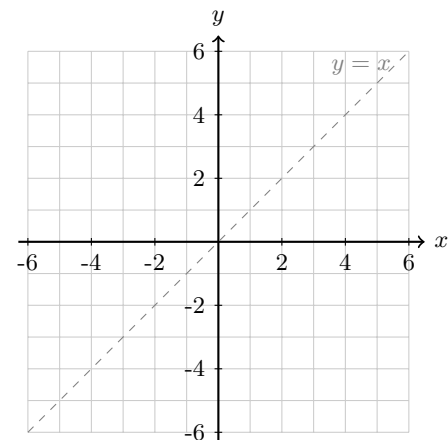
Part 1: Graphing Function and Inverse

For each: (a) find the inverse algebraically, (b) complete the point tables, (c) graph both on the grid.

1. $f(x) = (x + 3)^3$

$f^{-1}(x) = \underline{\hspace{2cm}}$

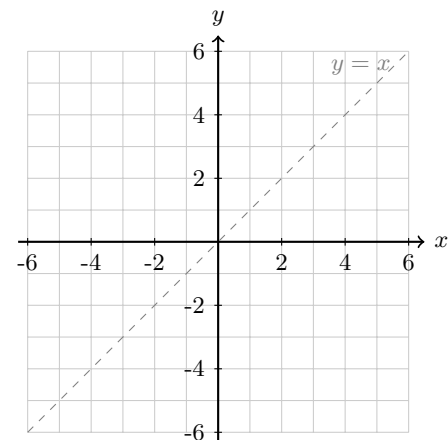
$f(x)$ points	$f^{-1}(x)$ points
(,)	(,)
(,)	(,)
(,)	(,)
(,)	(,)
(,)	(,)



2. $f(x) = \sqrt[3]{2x}$

$f^{-1}(x) = \underline{\hspace{2cm}}$

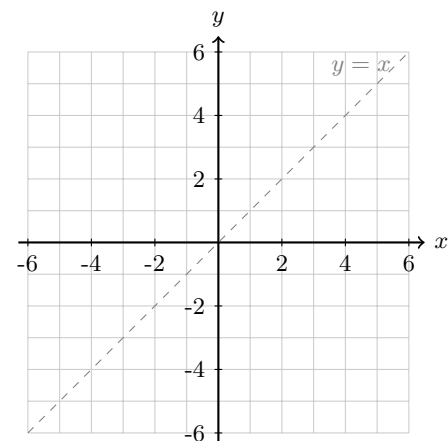
$f(x)$ points	$f^{-1}(x)$ points
(,)	(,)
(,)	(,)
(,)	(,)
(,)	(,)
(,)	(,)



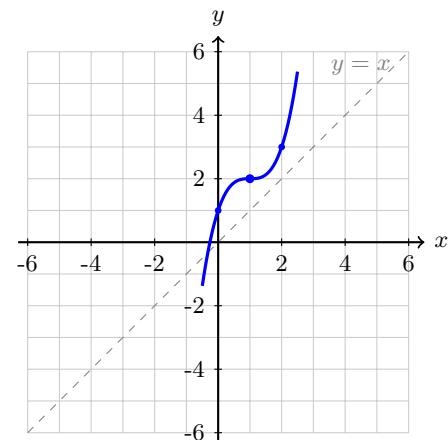
3. $f(x) = x^3 - 2$

$f^{-1}(x) = \underline{\hspace{2cm}}$

$f(x)$ points	$f^{-1}(x)$ points
(,)	(,)
(,)	(,)
(,)	(,)
(,)	(,)
(,)	(,)



4. Sketch $f^{-1}(x)$ for the shifted cubic below.

The inflection point of $f(x)$ is $(1, 2)$.Inflection point of $f^{-1}(x)$: _____

Part 2: Algebraic Mastery

Find the inverse of each function algebraically. Show your 4 steps clearly.

5. $f(x) = \sqrt[3]{5x+9}$

6. $f(x) = (x-5)^3 + 1$

$f^{-1}(x) = \underline{\hspace{2cm}}$

$f^{-1}(x) = \underline{\hspace{2cm}}$

7. $f(x) = \frac{1}{3}\sqrt[3]{x+7}$

8. $f(x) = 64x^3 + 3$

$f^{-1}(x) = \underline{\hspace{2cm}}$

$f^{-1}(x) = \underline{\hspace{2cm}}$

9. $f(x) = 216(x-2)^3$

10. $f(x) = \sqrt[3]{x+8} - 6$

$f^{-1}(x) = \underline{\hspace{2cm}}$

$f^{-1}(x) = \underline{\hspace{2cm}}$

11. $f(x) = -2(x+4)^3 - 5$

12. $f(x) = \sqrt[3]{3(x-1)} + 2$

$f^{-1}(x) = \underline{\hspace{2cm}}$

$f^{-1}(x) = \underline{\hspace{2cm}}$

Part 3: Short Answer

13. What line are a function and its inverse reflected across?

Answer: _____

14. If the point $(4, 64)$ is on $f(x) = x^3$, what point must be on $f^{-1}(x)$?

Answer: _____

15. Explain in your own words why swapping x and y in an equation gives you the inverse.

16. The functions $f(x) = 8x^3$ and $g(x) = (2x)^3$ produce identical graphs. Explain why algebraically.

Part 4: Application

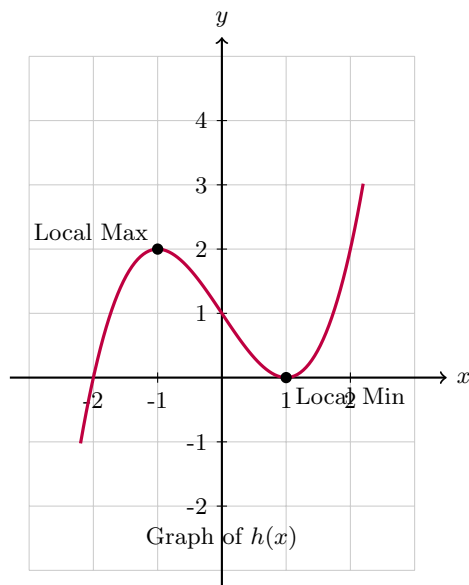
The volume of a cone where the radius equals the height is given by $V = \frac{1}{3}\pi h^3$.

17a. Solve for h in terms of V (find the inverse).

17b. Find h when $V = 2300 \text{ in}^3$. Round to the nearest tenth.

Part 5: Extension — Domain Restrictions

The graph below shows a polynomial function $h(x)$. Because this graph "turns," it fails the **Horizontal Line Test**, meaning it does not have an inverse function over its entire domain.



18. Identify three intervals for x where $h(x)$ would be one-to-one (invertible).

Interval 1: _____

Interval 2: _____

Interval 3: _____

19. On the middle section, is the function increasing or decreasing?

Answer: _____

20. Why does $f(x) = x^3$ not require a domain restriction to have an inverse, but $h(x)$ does?

Part 6: Challenge Problems

21. A cubic function has its inflection point at $(-2, 5)$ and passes through the point $(-1, 3)$. Write the equation of the function.

22. Find the inverse of $f(x) = \sqrt[3]{\frac{x-4}{2}} + 3$. Show all steps.

23. If $f(x) = (x+a)^3 + b$ and $f^{-1}(x) = \sqrt[3]{x-2} - 5$, find the values of a and b .

$a =$ _____

$b =$ _____

24. A cube root function has its inflection point at $(3, -1)$. If $f(11) = 1$, find the value of a and write the equation.

$a =$ _____

$f(x) =$ _____

25. Verify that $f(x) = 2x^3 + 5$ and $g(x) = \sqrt[3]{\frac{x-5}{2}}$ are inverses by showing that $f(g(x)) = x$.